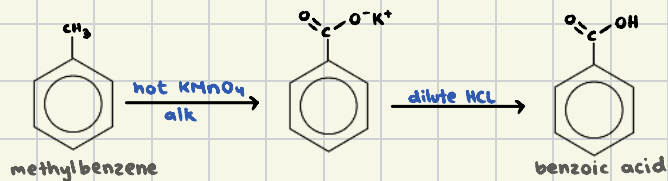


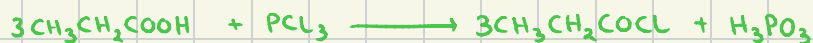
33 - Carboxylic acids & derivatives

Production of benzoic acid:



Reactions of carboxylic acids to form acyl chlorides

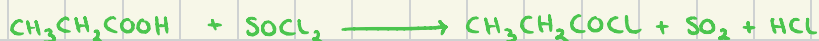
1) carboxylic acid + PCl_3 [& heat]



2) carboxylic acid + PCl_5

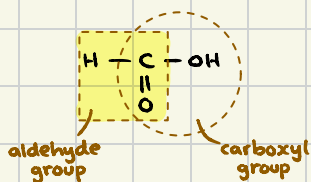


3) carboxylic acid + SOCl_2



Further oxidation of carboxylic acids $\leftarrow$ some carboxylic acids can be oxidised further

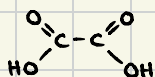
1) Methanoic acid



$\rightarrow$ the aldehyde-like hydrogen allows further oxidation to $\text{CO}_2 + \text{H}_2\text{O}$

* using oxidising agents like Fehling's reagent, Tollens' reagent, acidified KMnO_4 or acidified $\text{K}_2\text{Cr}_2\text{O}_7$

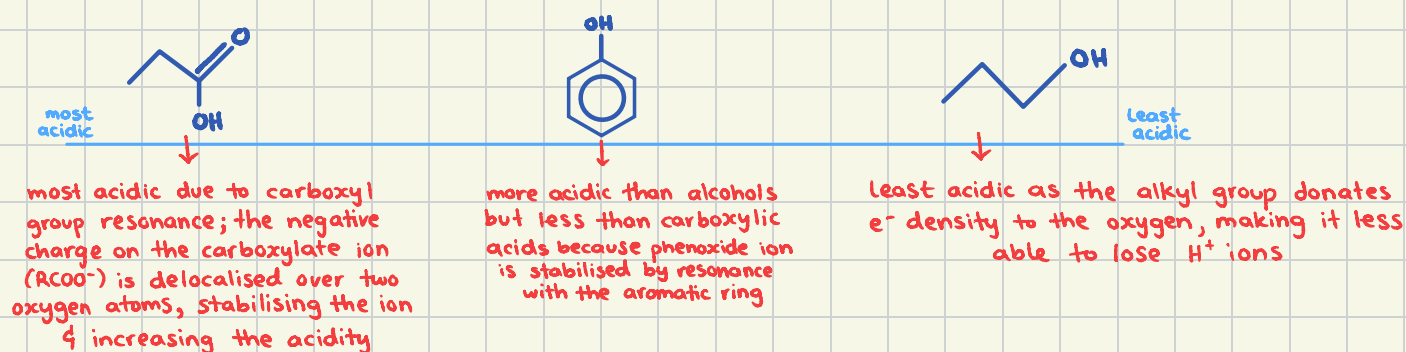
2) Ethanedioic acid



$\rightarrow$ can oxidise to form CO_2

* using warm acidified KMnO_4

Acidity of carboxylic acids VS phenols VS alcohols



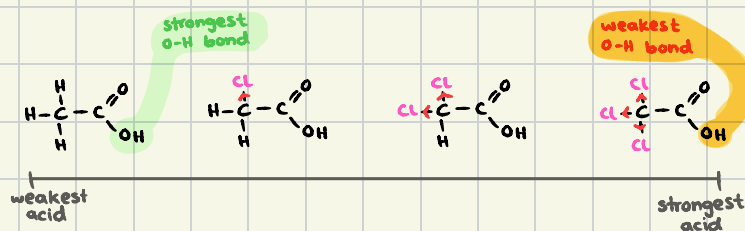
Acidity of chlorine-substituted carboxylic acids

→ chlorine-substituted carboxylic acids are MORE acidic than unsubstituted carboxylic acids

→ this is due to the electron-withdrawing effect of chlorine

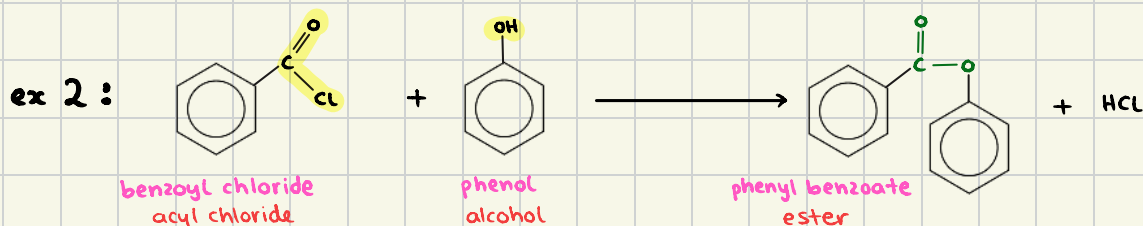
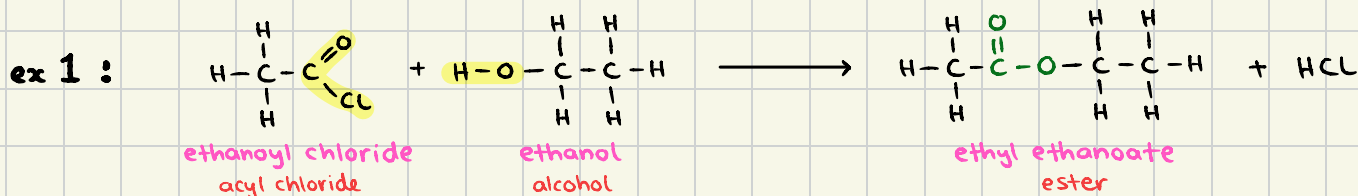
→ Cl, being electronegative, pulls e^- density away from the carboxyl group, stabilising the negative charge on the conjugate base (RCOO^-), & making it easier to lose the H^+ (thus more acidic)

→ as the number of Cl atoms increase, acidity increases



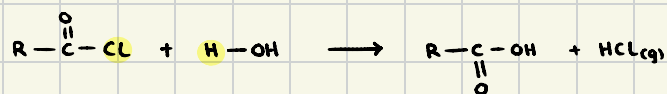
Production of esters

i) alcohol + acyl chloride



Reactions of acyl chlorides

1) reaction with water (hydrolysis)



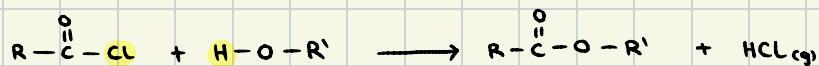
acyl chloride + water $\longrightarrow$ carboxylic acid + HCl

reagent: water

conditions: room temp.

products: carboxylic acid + HCl

2) reaction with alcohol (esterification)



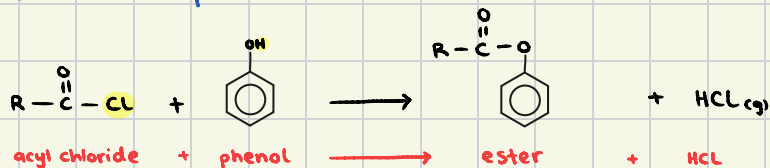
acyl chloride + alcohol $\longrightarrow$ ester + HCl

reagent: alcohol

conditions: room temp.

products: ester + HCl

3) reaction with phenol



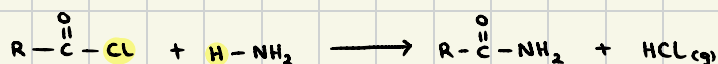
acyl chloride + phenol $\longrightarrow$ ester + HCl

reagent: phenol

conditions: room temp.

products: phenyl ester + HCl

4) reaction with ammonia



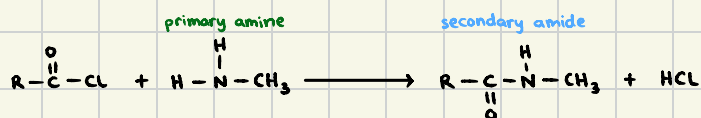
acyl chloride + ammonia $\longrightarrow$ amide + HCl

reagent: ammonia

conditions: room temp.

products: primary amide + HCl

5) reaction with amine

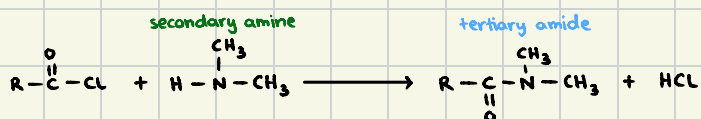


acyl chloride + primary amine $\longrightarrow$ secondary amide + HCl
(N-substituted amide)

reagent: primary amine

conditions: room temp.

products: N-substituted amide



acyl chloride + secondary amine $\longrightarrow$ tertiary amide + HCl
(N,N-substituted amide)

reagent: secondary amine

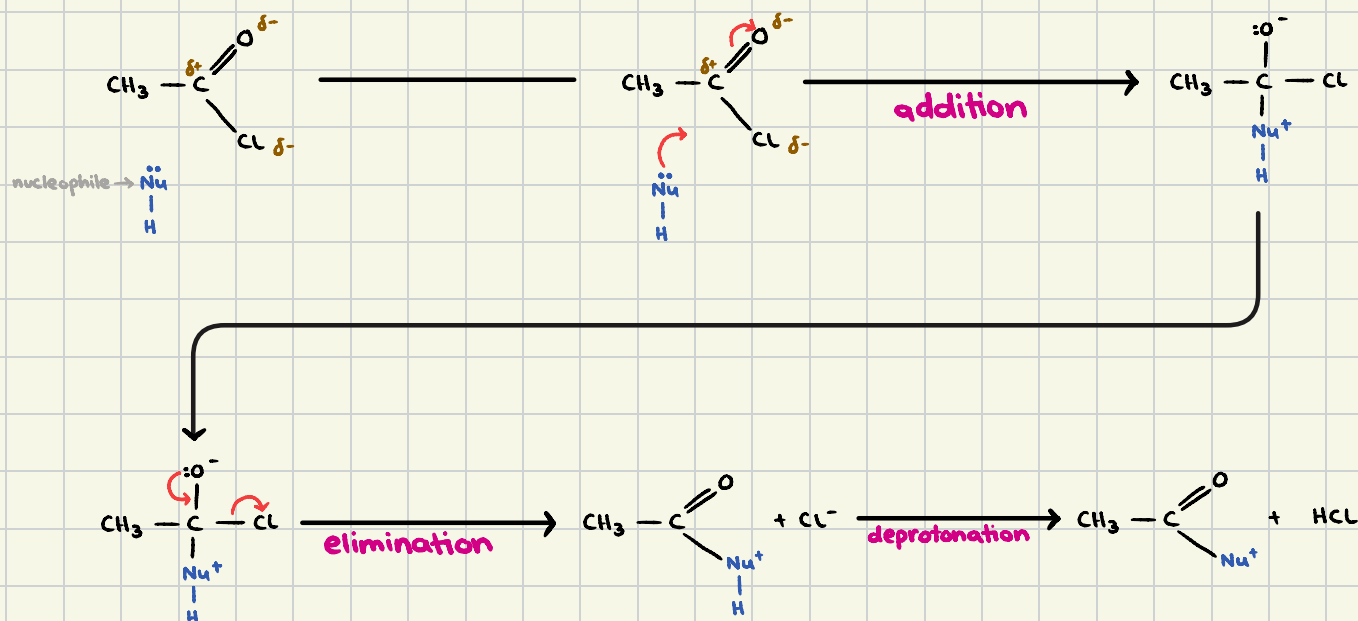
conditions: room temp.

products: N,N-substituted amide

formation of esters

formation of amides

Addition-elimination mechanism of acyl chlorides



Relative ease of hydrolysis: Acyl chlorides VS Alkyl chlorides VS Halogenoarenes

Acyl chlorides



easily hydrolysed

due to the high reactivity of the carbonyl carbon, which is strongly polarised by the carbonyl oxygen & the electronegative chlorine

this makes them highly susceptible to nucleophilic attack, even at room temp.

Alkyl chlorides



less reactive than acyl chlorides

they lack carbonyl group, which reduces the e^- density around the carbon attached to chlorine atom

this makes them require stronger conditions like heat & a strong base

Halogenoarenes



least reactive

due to the resonance stabilization of the C-Cl bond so the delocalised π system of the benzene ring reduces the partial +ve charge on the carbon attached to the chlorine

this makes them require extreme conditions like high temp. & pressure